

Ltd – an American-Israeli cybersecurity solution provider.

They predict that throughout 2025 and beyond, AI-powered tools will enable criminals to mimic the behaviour, appearance, and voice of individuals, creating realistic impersonations.

This is particularly dangerous on professional networks like LinkedIn, where the expectation of legitimate connections makes it easy for bad actors to infiltrate.

Criminals are increasingly using social engineering tactics, employing AI to craft convincing impersonations

and launch large-scale attacks. AI-driven bots and deepfakes can impersonate high-profile individuals, blurring the lines between reality and deception. The potential for these attacks to compromise corporate security and personal data is significant.

09 QUANTUM THREATS WILL DEBUT

Quantum computing is poised to revolutionise cybersecurity, introducing both unprecedented capabilities and significant threats. By 2025, advancements in quantum technology are expected to challenge traditional cryptographic defences, particularly public-key algorithms like RSA and ECC, which could be rendered vulnerable to quantum attacks. The US National Institute of Standards and Technology (NIST) has been proactive in addressing these concerns, releasing post-quantum cryptography standards in 2024.

However, the transition to these new standards is anticipated to be gradual, with larger enterprises, especially in finance, needing to begin planning for this quantum shift to protect sensitive data.

The concept of "harvest now, decrypt later" underscores the urgency of adopting quantum-resistant encryption. Adversaries may collect encrypted data now, anticipating future quantum capabilities to decrypt it, compromising long-term data security. In response, organisations are expected to invest



in quantum-safe encryption methods and develop comprehensive migration strategies to safeguard critical information. This proactive approach is essential to mitigate the looming quantum threat and ensure the integrity of digital communications in the quantum era.

The proliferation of Internet of Things (IoT) devices is anticipated to significantly expand the cyber threat landscape through 2025 and beyond. As IoT adoption accelerates, the number of connected devices is projected to reach 21.5 billion by 2025, substantially increasing potential attack surfaces. Many IoT devices lack robust security measures, making them susceptible to exploitation. Cybercriminals are expected to target these vulnerabilities to launch large-scale Distributed Denial of Service (DDoS) attacks, infiltrate networks, and exfiltrate sensitive data.

The integration of IoT in critical infrastructure sectors, such as healthcare, energy, and transportation, introduces additional risks. Compromises in these

10 DIVERSIFIED IOT ATTACKS

areas could lead to severe disruptions and safety hazards. For instance, vulnerabilities in medical IoT devices may jeopardise patient safety, while attacks on smart grids could disrupt power



supplies. To mitigate these evolving threats, organisations must implement comprehensive security strategies, including device hardening, secure communication protocols, and continuous monitoring. Regulatory bodies are also expected to enforce stricter cybersecurity standards for IoT devices to foster a more secure ecosystem. In summary, the rapid expansion of IoT presents significant cybersecurity challenges that necessitate proactive measures to safeguard against emerging threats.